Brightwater

Technical assessment submission

The Meraki Group | 26 October 2026

Website, proposed telephony design, support-ticket response and development summary.

Live website: Open Brightwater

Source and README: GitHub repository

Staff workspace: Operations

Contents

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Acceptance and remaining items

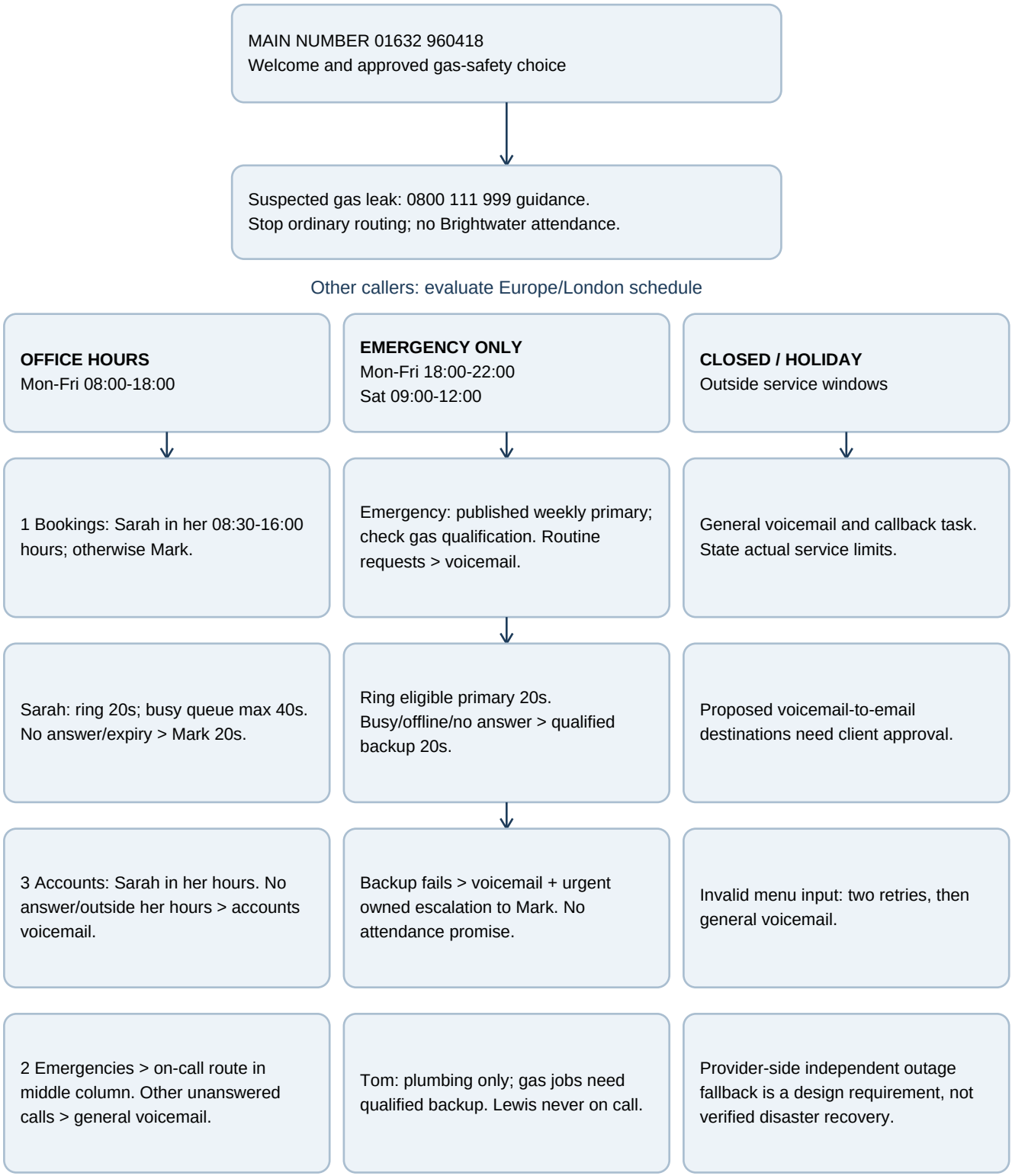
The owner confirmed successful website and call testing on a physical mobile device on 26 October 2026. Detailed device and per-scenario records were not supplied.

55 automated tests passed at the helpdesk release. The public site and workspace are deployed on Railway. Free-hosting compliance and original Brand Pack assets remain unresolved. This PDF is prepared for submission; it has not been emailed to Meraki.

Assessment prototype / supervised pilot. No enterprise certification is claimed. All client configuration and email drafts below are proposals. The live Plivo and Gmail extensions do not imply that the client BT migration has occurred.

Task 2 | Call-flow diagram

Proposed client routing; not a record of an implemented BT migration.



Task 2 | Build sheet and client communication

Person	Extension	Endpoint	Policy
Sarah Collins	201	Windows softphone + USB headset; optional supported desk phone	Bookings/accounts, weekdays 08:30-16:00
Mark Ellis	202	Supported iPad/mobile app	Bookings fallback; owner escalation; Gas Safe
Dan Hughes	203	Supported personal-mobile work app	On-call rota; Gas Safe
Priya Shah	204	Supported personal-mobile work app	On-call rota; Gas Safe
Tom Barker	205	Supported personal-mobile work app	On-call plumbing only; gas jobs to qualified backup
Lewis Grant	206	Optional supervised app, license subject to approval	Apprentice; never emergency on-call

Extensions are proposed, not existing facts. Confirm app/OS support, licenses, BYOD consent, battery permissions, mobile reception and headset needs. Do not assume the three existing BT cordless handsets support hosted SIP; inventory models first.

Object	Proposed setting
Main number	Port 01632 960418 only after account ownership, service association and port eligibility checks
Secondary number	01632 960419: do not cancel or port until its use, alarms/fax/dependencies and ownership are confirmed
Schedule	Europe/London with daylight-saving transitions; office and emergency windows as above
Holidays	Explicit approved dates; simulator closes the whole day; agree any live emergency exception separately
Welcome	Short brand welcome, then 1 bookings / 2 emergencies / 3 accounts
Invalid input	Two retries then general voicemail; verify provider timeout/retry semantics
Bookings group	Sarah, then Mark; Sarah omitted outside her working hours
Busy handling	Queue max 40 seconds; offer voicemail; expiry proceeds to fallback; confirm concurrency/licenses
Accounts	Sarah only during her hours; otherwise accounts voicemail
Emergency group	Current weekly primary, then qualified backup; no Lewis; Tom requires gas-qualified escalation
Ring timeout	20 seconds per destination, configurable 10-60 seconds in demo
Answer confirmation	Live provider should require engineer acceptance so personal carrier voicemail cannot consume escalation
General voicemail	Proposed info@brightwaterplumbing.co.uk, owned callback queue

Object	Proposed setting
Accounts voicemail	Proposed accounts@brightwaterplumbing.co.uk
Emergency failure	Urgent follow-up owned by Mark, with confirmed independent notification and acknowledgment policy
Outbound identity	Verified/authorized main number via provider app or supported callback; native mobile calls retain mobile identity
Call recording	Off by default; voicemail only until purpose, access, notice and retention are agreed
Outage	Provider-hosted announcement/fallback independent of website, plus alternate connectivity and documented escalation

Mailboxes are proposed destinations derived from the supplied address prefixes; confirm full domain, recipients, ownership and delivery controls. These client voicemail destinations remain proposals. Separately, the implemented helpdesk supports explicitly selected Gmail ticket replies; it does not activate these proposed voicemail destinations.

Questions before live configuration

1. Which Meraki platform, license package and mobile apps will be used? This determines queue, failover, acceptance and caller-ID capabilities.

2. Who covers the 08:00-08:30 and 16:00-18:00 gaps, accounts absences, lunches and leave? Demo assumes Mark for bookings and voicemail for accounts.

3. Who is each week's backup, who publishes the rota, and how do both acknowledge handover? Confirm qualified cover when Tom is primary and what happens if the backup cannot attend.

4. Who receives urgent alerts, by which independent channel, and who acts when no acknowledgment arrives? Agree a realistic threshold before configuring it.

5. Are the 20-second rings and 40-second queue acceptable? Confirm simultaneous calls, waiting prompts and voicemail escape key.

6. What happens on bank holidays and planned closures? Demo's full-day closure is only a proposal.

7. Confirm both BT numbers, account holder, billing address, contract/port details and the second number's purpose. Broadband is on the main line and ends March 2027; porting may affect connectivity or fees.

8. Confirm mailbox addresses, access, retention, consent/notice and whether recordings or transcriptions are wanted. Never email sensitive recordings to an unconfirmed group.

9. Confirm handset models, engineer mobile OS/devices, reception, app permissions, Wi-Fi and mobile data; agree BYOD/work-hours arrangements.

10. Confirm the port window, temporary number, fallback contacts, provider rollback options and acceptance-test sign-off. Do not cancel BT before a coordinated migration.

Draft email to Sarah - not sent

Hi Sarah,

The proposed system gives callers three choices: bookings, emergencies and accounts. Bookings reach you during your working hours, then Mark if you cannot answer. A short queue will help when you're already on a call. Accounts reach you or their own voicemail.

Emergency calls go to that week's engineer, then an agreed backup. If neither answers, Mark gets an urgent follow-up. Tom's gas-related jobs need a Gas Safe engineer. Outside the agreed service hours callers can leave a message, with clear guidance that emergency attendance is unavailable.

Engineers will use the phone-system app to call with the office number. Calling from the ordinary mobile dialer will still show their personal number.

Before setup, please confirm who covers your early/late hours and holidays; who approves and backs up the weekly rota; and who receives voicemail and urgent alerts. We'll also need the BT account details and the purpose of the old fax number. Because broadband shares the main line, we'll agree the migration with BT and our provider before moving anything.

We'll test the new system on a test number first and agree the transfer date with you.

Thanks

Task 3 | Investigation of ticket #10482

Treat Saturday emergency failure as high priority and Mark's caller ID as a linked, separate work item. Neither is marked fixed without evidence. The simulator reproduces possible routes; it does not establish the actual cause.

1. Obtain the three call times, caller number securely, number dialed and timezone; confirm whether the customer still needs help through the client's established process. Do not promise an engineer or a same-day technical repair.

2. Match provider inbound logs and call IDs. Verify calls reached the expected number/tenant, not an old answerphone or redirect.

3. Inspect Saturday 09:00-12:00 schedule, timezone, holidays and the effective configuration version. A wrong schedule could send callers to voicemail before the rota is read.

4. Compare published rota, acknowledgment and audit history to Priya's intended assignment. Check whether routing used a stale group, wrong extension, missing backup or unacknowledged handover.

5. Inspect each ring leg: destination, offer time, response, timeout and final disposition. Distinguish an unoffered call from a device that never notified, and provider voicemail from personal carrier voicemail.

6. With Priya, check registration, app sign-in, push/notification permissions, Focus/DND, battery restrictions, network/data reception and app version. Check business hours/presence on the provider as well as the handset.

7. Inspect backup route, queue overflow, voicemail destination and alerts. Confirm Mark actually received and acknowledged the fallback task; an emailed voicemail alone is insufficient for an urgent route.

8. Reproduce on a test number with Priya and the backup: answered, declined, busy, offline, no-answer and carrier-voicemail cases. Record timestamps and call IDs. Change one cause at a time and retain the previous configuration.

9. For Mark separately, establish whether he used the provider app or native dialer. Check verified outbound CLI, permissions, number assignment and provider/carrier restrictions, then test to an external phone. Never suggest arbitrary spoofing.

Information needed: exact times and numbers; Priya's expected rota, device/OS/app, network and presence; Mark's calling method and example destination/time; relevant call IDs, effective configuration, recent changes and alert delivery evidence. Collect personal data through a secure support channel.

Task 3 | Initial response and prevention

Initial reply, within the hour - draft, not sent:

Hi Sarah, I'm sorry those emergency calls did not reach Priya. I'm treating this as high priority and checking the Saturday call records, rota and Priya's app together. Please send the call times and the number the customer dialed through our secure support channel, and let us know whether the customer still needs assistance. I'll agree a temporary cover route with you while we investigate and update you within an hour, even if the cause is still being checked. I have also logged Mark's caller-ID issue separately so it is not lost behind the emergency fault. We will test both fixes with you before closing them.

Prevention after diagnosis: acknowledged primary/backup handover; Friday test call; qualified backup coverage; alerts for unregistered destinations and unanswered emergencies; owned follow-ups with acknowledgment; versioned configuration and rollback; scheduled test calls for Saturday/holiday/DST boundaries; app onboarding and BYOD guidance. Add tests for the actual root cause once known, rather than declaring every possible failure fixed.

Task 4 | Written summary (1 of 2)

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Approach and priorities

The starting priority was a usable public website with accurate prices, coverage restrictions, emergency guidance and a straightforward enquiry journey. Next.js and TypeScript were selected with the user. The scope then expanded at the user's request to staff accounts, operations, automations and simulated AI calls. These additions do not replace the assessment's written telephony/support deliverables.

The site has separate Home, Services, Pricing, Coverage, About, Contact and Landlords pages. Enquiries persist with a reference for office follow-up. The workspace provides role-based access, assignment, internal tasks, rota, audit events and record-based analytics. Bookings, dispatch and general customer messaging remain simulated. An inbound telephone pilot carries real audio; the helpdesk now supports live Gmail ticket replies, isolated-folder intake, private portal links and overdue tasks.

Decisions and assumptions

One Railway web service serves both website and workspace; PostgreSQL, Redis, worker and beat are separate services. SQLite remains a convenient local-development fallback. Business records are authoritative; Redis schedules idempotent work. DeepSeek uses allowlisted tools for business facts and proposes actions rather than executing customer deliveries. ElevenLabs provides browser speech/transcription and bidirectional telephone audio through a Plivo WebSocket bridge, with interruption handling. Human consent and operator approval remain distinct. The agent identifies as AI and escalates uncertainty.

The assessment's fictional people, pricing, postcode rules and gas-emergency number are treated as requirements. Missing Brand Pack assets are replaced by clearly labelled illustrative imagery and a temporary wordmark. Telephony platform details and operational choices remain proposed assumptions in the build sheet, with client questions rather than invented confirmations.

Quality checks and lessons

Routing/coverage, permissions, session revocation, task ownership, version conflicts, automation idempotency, AI tool limits and voice ownership have automated coverage. Browser checks covered desktop/mobile viewports and key workspace flows. Earlier live provider checks confirmed speech/transcription and selected AI tool cases. Database migration and isolated snapshot restore were verified. The owner confirmed successful website and call testing on a physical mobile device on 26 October. This is user-reported acceptance, not independent load, accessibility or security certification.

Problems found included local dependency-read stalls, legitimate requests rejected by origin handling, hard-coded engineer lists, microphone access disabled by a security header, and a queued drag update reading cleared state. These were corrected. The map now limits updates to animation frames and passed a synthetic 900-gesture/9,000-movement stress test. Passing tests does not establish universal device compatibility or enterprise reliability.

Task 4 | Written summary (2 of 2)

Tools and AI assistance

ChatGPT/Codex provided substantial help with requirements analysis, design exploration, code, debugging, tests and documentation. Recorded design workflows included Taste Skill and Impeccable; gstack browser tooling supported verification. AI image generation produced concepts and labelled illustrative photography. Next.js, React, TypeScript, Node.js, PostgreSQL, SQLite, BullMQ/Redis, Docker, Railway, DeepSeek and ElevenLabs form the implementation stack.

AI output was not accepted as proof: generated mockups needed factual correction, and browser integration exposed issues that unit tests missed. The distinction between a simulated event, provider response and real customer outcome is essential. The latest voice adjustment allows a longer pause before transcription and adds concise, calm handling guidance for frustrated or abusive callers; broad live conversational evaluation remains outstanding.

Outstanding handover

Receive the Brand Pack, resolve the brief's free-hosting requirement, formally deliver the public repository link to Meraki, and send the packaged PDF assessment submission. Real use would additionally require independent recovery, security/accessibility/load review, stronger identity lifecycle and broader monitored delivery validation. No real number port, customer email or submission email has been sent. The accompanying checklist records these gaps openly.